

Database Systems Masterclass

DBMS Architecture, Relational Model, SQL, Normalization & Transactions

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Module 1: Introduction to Database Systems

1. What is a Database System?

- **Definition:** An organized collection of structured information or data stored electronically in a computer system.
- **DBMS (Database Management System):** Software package enabling users to define, create, maintain, and control database access.
- **Key Objective:** Eliminate data redundancy, prevent inconsistency, ensure security, and handle concurrent access efficiently.

2. Characteristics & ANSI/SPARC Architecture

- **Self-Describing Nature:** Contains stored database data alongside complete metadata definitions and constraints.
- **Program-Data Independence:** Allows changing data storage structures without modifying application programs.
- **Three-Schema Architecture:**
 - *Internal Level:* Physical storage structures and access paths.
 - *Conceptual Level:* Logical schema describing overall data and relationships.
 - *External Level:* User views customized for specific application groups.

Module 2: The Relational Model & SQL

1. Relational Model Fundamentals

- **Relation (Table):** Matrix of rows and columns introduced by E.F. Codd (1970).
- **Tuple & Attribute:** Tuple represents a single record row; attribute represents a named column property.
- **Integrity Constraints:**
 - *Entity Integrity:* Primary key attributes cannot be null.
 - *Referential Integrity:* Foreign keys must match a valid primary key or be null.

2. SQL Sub-languages

Sub-language	Description & Key Commands
DDL	Data Definition Language (CREATE, ALTER, DROP)
DML	Data Manipulation Language (SELECT, INSERT, UPDATE, DELETE)
DCL	Data Control Language (GRANT, REVOKE security rights)
TCL	Transaction Control Language (COMMIT, ROLLBACK)

Module 3: Database Design & Normalization

1. Entity-Relationship (ER) Modeling

- **Core Components:** Entities (objects), Attributes (properties), Relationships (associations), and Cardinality Ratios (1:1, 1:N, M:N).

2. Database Normalization

The systematic process of organizing tables to minimize data redundancy and dependency anomalies.

- **1NF (First Normal Form):** Eliminates repeating groups; ensures atomic attribute values.
- **2NF (Second Normal Form):** Meets 1NF and removes partial dependencies on composite keys.
- **3NF (Third Normal Form):** Meets 2NF and eliminates transitive non-key dependencies.
- **BCNF:** Stricter version of 3NF ensuring every determinant is a candidate key.

Module 4: Transactions, Concurrency & Recovery

- **ACID Properties:** Atomicity (all-or-nothing), Consistency (valid state transitions), Isolation (independent execution), and Durability (persistent commits).
- **Concurrency Control:** Two-Phase Locking (2PL) protocols preventing lost updates and dirty reads.

Module 5: Advanced Topics & NoSQL

- **Indexes & Optimization:** B-Trees and Hash indexes speeding up data lookup; query optimizers select cost-effective execution paths.
- **NoSQL Databases:** Designed for horizontal scaling and unstructured big data:
 - *Document Stores:* MongoDB, CouchDB
 - *Key-Value Stores:* Redis, DynamoDB
 - *Column-Family & Graphs:* Cassandra, Neo4j